Problem Set 33

Integration by Parts

1 Calculate the following integrals using integration by parts.

(a) $\int \arctan(x)\,dx$

(b) $\int x\cos(x)\,dx$

(c) $\int (\ln x)^2\,dx$

(d) $\int e^x \sin(x)\,dx$

(e) $\int x^2 \sin(x)\,dx$

2 Calculate the following integrals by using an appropriate strategy.

(a) $\int x\cos(x^2)\,dx$.

(b) $\int_1^2 \frac{2xe^{x^2}}{e^{x^2+5}}\,dx$

(c) $\int x\ln\left(x^2+1\right)\,dx$.

(d) $\int \frac{3x^2}{x^2+2x+1}\,dx$.

(e) $\int x^3 e^{x^2}\,dx$.

(f) $\int -\frac{\ln(1+x)}{(1+x)^2}\,dx$.

3 (Extra Credit, 2 points): The math department is interested in better understanding your math experience this semester. You can help by filling out this survey, which should take around 10-15 minutes to complete. Upon completing the survey you will have the option to opt-in to allow your responses to be included in a broader research study Harvard is involved in to understand the experiences of university students in introductory math courses.

We do need you to fill in your name and HUID in order to receive the extra credit for the survey. Your responses will be aggregated by a research team and anonymized before being shared with the teaching team.